



Summer Internship Project Report, 2024

Understanding the Connection between Random Walk and Heat Equation



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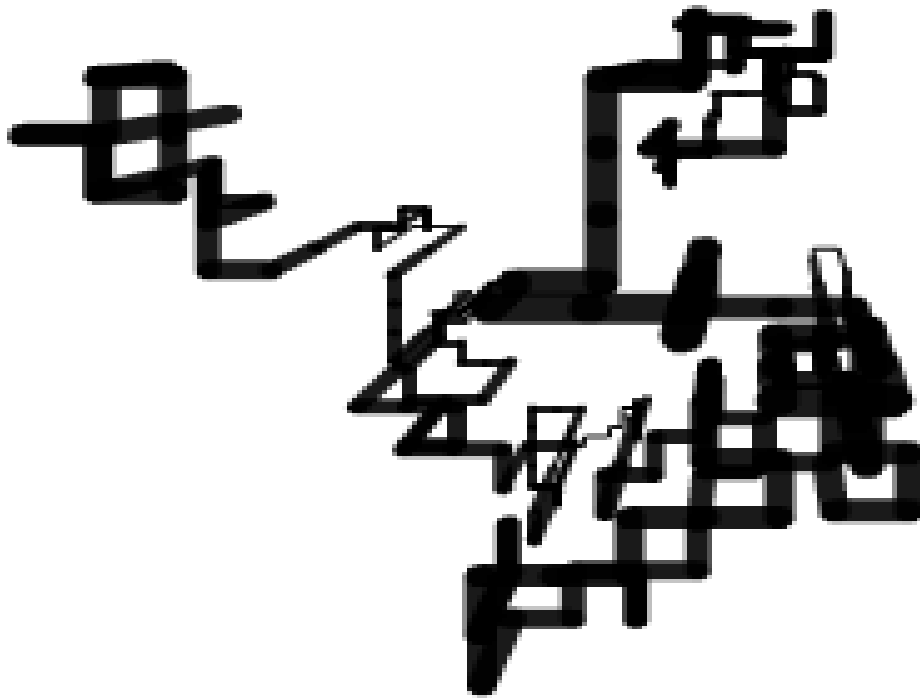
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Sincerely,
Arpan Saha

Title of Research Project

*Understanding the Connection between Random
Walk and Heat Equation*



Objectives

Random Walk is a very important concept in probability theory that is used in many contexts such as framing the concept of Brownian motion and heat equation in physics. The applications of the concept of random walk also range over many disciplines like chemistry, materials science, biology, economics etc. The primary objectives of this project work are described in the following.

Understanding the concept of random walk

To study the connection between random walk and heat equation, it is very important to understand the fundamental theory of random walk. In this project, the aim is to study this concept and understand the connection with the heat equation.

Exploring the heat equation from a different perspective

To know the relation between random walk and heat equation, it is very important to see the heat equation from a different perspective. This project aims to cover the basic concepts of heat equation and then explore it in a different way.

Having these objectives, this project on random walk and heat equation explains a very crucial bridge between these two concepts and thereby sheds some light on the interconnections between the disciplines of probability theory and differential equations.

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Chapter 1

Introduction of Research Topic

The concepts of random walk and heat equation are among the most fascinating topics in mathematics and understanding the concept of connection between them is like making a bridge between the two topics and thereby creating a wonderful intersection of the two subjects, probability theory and partial differential equations. In this project report, first, we focus on some basic concepts of probability theory. Then, we give an introduction to the concept of random walk in one dimension, and then we gradually uplift the idea to the higher dimensions. In the other part, we introduce the concept of the heat equation in a general way i.e. the deterministic point of view. The heat equation is a deterministic (non-random), partial differential equation derived from this intuition by averaging over a very large number of particles. This equation can and has traditionally been studied as a deterministic equation. But we study heat equation in discrete form, from a probabilistic point of view which will add an extra layer of enrichment to this concept and finally, we will shed some light on the connection between random walk and the discrete heat equation.

Chapter 2

Theoretical Framework of Topic

The study of understanding the connection between random walk and heat equation is one of the most fascinating concepts in mathematics. To explore this area properly, it is very essential to have a good grip on the basic concepts. This section covers the basic concepts related to this work in detail and thereby provides a strong foundation to this topic.

2.1 Basic concepts from probability theory

Probability theory is a branch of mathematics, concerned with random phenomena. The basic concepts of probability theory that provide the whole framework of this project work are discussed in the following:

Notes about probability

To know about probability, first, we have to know about sample space and events. A sample space is a set Ω and an event is a collection, $\mathcal{F}$ of subsets of Ω including ϕ and Ω satisfying the conditions:

- i) If $A (\subset \Omega) \in \mathcal{F}$, then $A^c \in \mathcal{F}$.
- ii) If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

Probability is a function from $\mathcal{F}$ to $[0, 1]$, i.e. $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$, satisfying the following conditions:

- i) $\mathbb{P}(\Omega) = 1, \mathbb{P}(\phi) = 0$.
- ii) If $E_1, E_2, \dots$ are mutually exclusive events, then

$$\mathbb{P} \left(\bigcup_{n=1}^{\infty} E_n \right) = \sum_{n=1}^{\infty} \mathbb{P}(E_n).$$

Conditional probability

If V_1 and V_2 are two events, then the conditional probability of V_1 given V_2 is given by

$$\mathbb{P} (V_1|V_2) = \frac{\mathbb{P}(V_1 \cap V_2)}{\mathbb{P}(V_2)},$$

provided $\mathbb{P} (V_2) > 0$. If X and Y are two discrete random variables, then the conditional probability mass function of X given $Y = y$ is given by

$$\mathbb{P}(X = x|Y = y) = \frac{\mathbb{P}(X=x, Y=y)}{\mathbb{P}(Y=y)},$$

provided $\mathbb{P}(Y = y) > 0$.

Random variable & random vector

A random variable is a function from the sample space to the set of real numbers. i.e. $X : \Omega \rightarrow \mathbb{R}$. If $X(\Omega)$ is at most countable, it is called discrete random variable. Otherwise, it is called continuous random variable.

$\mathbf{X} = (X_1, X_2, \dots, X_n)$ is an n dimensional random vector if all its components are real-valued random variables. i.e. $\mathbf{X} : \Omega \rightarrow \mathbb{R}^n$.

Expectation and variance of a random variable

The expectation and variance of a discrete random variable X are defined respectively as

$$\begin{aligned}\mathbb{E}[X] &= \sum_z z \mathbb{P}\{X = z\}, \\ \text{Var}[X] &= \sum_z z^2 \mathbb{P}\{X = z\}.\end{aligned}$$

2.2 Random walk

It is a random process that describes a path that consists of a succession of random steps on some mathematical space. Here, we consider the walk of a person on the set of integers on the real line in one dimension with equal probability. We can also uplift the idea to higher dimensions. In higher dimensions, the walker walks on the set $\mathbb{Z}^d = \{(x_1, x_2, \dots, x_d) | x_i \in \mathbb{Z}\}$ and the walker moves to $2d$ possible directions from a particular position.

2.3 Heat equation

We study the heat equation,

$$u_t(t, x) = c \Delta u(t, x), \quad \text{where } c > 0.$$

Here, $u : \mathbb{R}^+ \times U \rightarrow \mathbb{R}$, where U is an open bounded set in $\mathbb{R}^d$. We have initial condition,

$$u(0, x) = f(x), \text{ where } x \in U,$$

and we have a boundary condition,

$$u(t, x) = 0, \text{ where } x \in \partial U.$$

The heat equation is a deterministic (i.e., without randomness) model for heat flow. It can be studied without probability. However, probability adds a layer of richness in terms of movements of individual random particles. This point of view is often useful for understanding the equation. We introduce the heat equation in a different perspective in the upcoming chapters of this report.

Chapter 3

Profile of Organization

National Institute of Science Education and Research (NISER), Bhubaneswar, is a unique institution in India, committed to the generation, dissemination and preservation of knowledge in basic sciences, integrated with the goal to create a zeal and fervor in minds of the young buds of the nation towards scientific temper. The Institute is affiliated with Homi Bhabha National Institute (HBNI), Mumbai, a deemed research university that oversees academic programs at Institutions that are part of the Department of Atomic Energy (DAE) structure.

3.1 Vision

NISER recognizes that modern scientific research is carried out in a domain without any boundaries. This entails encouraging a new scientific culture where members of our community attain to an intellectual agility unconstrained by the limitations of disciplinary conventions from the past. The students and research scholars at NISER flourish in the ambiance of this scientific culture. In this perspective, the faculty members from different schools try to establish centres of research in interdisciplinary areas while discharging their teaching duties of core courses of the respective schools.

3.2 Research

Mathematics occupies a core intellectual position at NISER, an institution that is striving to be recognized as a Center of Excellence in science education and research in basic sciences. The School of Mathematics was established as a core department of the NISER along with other Schools from the very beginning of NISER in 2007. Along with teaching, research is a main goal of NISER. The School of Mathematics aims to have a vibrant academic and research ambience for faculties, students, and visitors. Currently, the NISER library houses about 10-20 journals in hard copy and has access to more than 100 mathematics journals including MathSciNet, ScienceDirect, and JSTOR. Currently, the faculties are experts in the research area of functional analysis, harmonic analysis, graph theory, probability theory, statistics, number theory, incident geometry, differential equations and cryptology.

Chapter 4

Methodology Followed

To pursue an internship project on any topic, there should be an organised methodology followed by the student, otherwise, the work will not be completed properly. In this chapter, we will discuss regarding the methodology followed for this project work.

4.1 Theoretical fundamentals

As a first step towards this project work, a thorough understanding of the topic was done with the help of various books (mentioned in the *References* section). This step includes various sub-steps which include studying the key concepts, theories, their applications, uplifting the ideas etc.

4.2 Daily presentations

To ensure the daily progress of the project work, continuous presentations were given to the supervisor regarding the summarization of the work completed on each day. These presentations include discussing the key topics, finding the problems, correcting the mathematical arguments, highlighting any challenges encountered and discussing potential solutions.

4.3 Preparing project report

After discussing all the issues related to the topic, a project report is prepared maintaining the prescribed format given by INSPIRE authority. After reviewing and correcting it multiple times, this report is submitted to the respective institute and INSPIRE portal.

Chapter 5

Analysis and Interpretation of Research Project

5.1 Random walk in one dimension

Consider the random walk on the integers. At each time unit, the walker flips the coin in the air and moves to the right or left according as it results head or tail. Let S_n be the position of the walker at time n . Let us assume that the walker starts walking at x . Then we can write,

$$S_n = x + X_1 + X_2 + \dots + X_n,$$

where $X_j = \pm 1 \forall j \in \{1, 2, \dots, n\}$, denotes the change of the position of the walker between time $j - 1$ and j . Note that, $X_1, X_2, \dots$ are independent random variables with $\mathbb{P}\{X_j = 1\} = \mathbb{P}\{X_j = -1\} = \frac{1}{2}$.

Expectation

Let us consider $x = 0$. Then by linearity property of expectation.

$$\mathbb{E}[S_n] = \sum_{i=1}^n \mathbb{E}[X_i].$$

Now, $\mathbb{E}[X_i] = 0$ since steps of $+1$ and -1 are equally likely. Therefore we can write,

$$\mathbb{E}[S_n] = 0.$$

Variance

If $X_1, X_2, \dots, X_n$ are independent random variables,

$$\text{Var}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \text{Var}[X_i].$$

Now, $\text{Var}[X_i] = 1 \forall i \in \{1, 2, \dots, n\}$. So,

$$\text{Var}[S_n] = n.$$

The probability of being at a particular position

If the walker starts journey at origin, it is easy to see that if he wants to be at the origin after some steps, he has to take even number of steps, $2n$. Out of these steps, he has to take n “+1” steps and n “−1” steps. Therefore,

$$\mathbb{P}\{S_{2n} = 0\} = \binom{2n}{n} 2^{-2n} = \frac{(2n)!}{n!n!} 2^{-2n}.$$

More generally, if a walker is to be at $2j$, there must be $(n + j)$ steps of +1 and $(n - j)$ steps of −1. Therefore,

$$\mathbb{P}\{S_{2n} = 2j\} = \binom{2n}{n+j} 2^{-2n} = \frac{(2n)!}{(n+j)!(n-j)!} 2^{-2n}.$$

Now, trying to understand the expression on the right hand side of the second equality leads to studying the behaviour of $n!$ as n gets large. We will discuss this in the following.

Stirling’s formula

It states that as $n \rightarrow \infty$,

$$n! \sim \sqrt{2\pi} n^{n+\frac{1}{2}} e^{-n},$$

where $\sim$ means that the ratio of the two sides tends to 1.

Proof. First we will prove that there is a positive number C_0 such that

$$\lim_{n \rightarrow \infty} b_n = C_0, \quad \text{where } b_n = \frac{n!}{n^{n+\frac{1}{2}} e^{-n}},$$

and then we will show that $C_0 = \sqrt{2\pi}$. Now, consider

$$b_n = b_1 \prod_{j=2}^n \frac{b_j}{b_{j-1}} = b_1 \prod_{j=2}^n [1 + \delta_j], \quad \text{where } \delta_j = \frac{b_j}{b_{j-1}} - 1,$$

and after taking log on both sides and taking limits we get,

$$\lim_{n \rightarrow \infty} \log b_n = \log b_1 + \sum_{j=2}^{\infty} \log [1 + \delta_j].$$

Now, by Taylor’s series expansion of the logarithm function, we know that

$$|\log[1 + \delta_j]| = \left| \delta_j - \frac{\delta_j^2}{2} + \frac{\delta_j^3}{3} - \dots \right|,$$

From this, we can say that

$$|\log[1 + \delta_j]| \leq |\delta_j| + \frac{|\delta_j|^2}{2} + \frac{|\delta_j|^3}{3} + \dots,$$

for $|\delta_j| \leq \frac{1}{2}$, the series can be expressed as,

$$|\log[1 + \delta_j]| \leq |\delta_j| \left(1 + \frac{|\delta_j|}{2} + \frac{|\delta_j|^2}{3} + \dots \right),$$

so, Taylor’s series for the logarithm shows that $|\log [1 + \delta_j]| \leq c |\delta_j|$ for some $c \in \mathbb{R}$ and for $|\delta_j| \leq \frac{1}{2}$ by comparison test. So the required sufficient condition for convergence of the sum is

$$\sum_{n=2}^{\infty} |\delta_n| < \infty.$$

Now it is easy to see that $b_1 = e$, and if $n \geq 2$,

$$\frac{b_n}{b_{n-1}} = e \left(1 - \frac{1}{n}\right)^n \left(1 - \frac{1}{n}\right)^{-\frac{1}{2}}.$$

Taking log on both sides and after Taylor's series expansion, we have

$$\log\left(\frac{b_n}{b_{n-1}}\right) = O(n^{-2}).$$

so, there exists $d \in \mathbb{R}$ such that

$$\log\left(\frac{b_n}{b_{n-1}}\right) \leq \frac{d}{n^2},$$

we know that $\delta_n = \frac{b_n}{b_{n-1}} - 1$. From this, we can say that

$$\begin{aligned} \delta_n &\leq e^{\frac{d}{n^2}} - 1 \\ \implies \delta_n &\leq 1 + \frac{d}{n^2} + \frac{d^2}{2n^4} + \dots - 1 \\ \implies \delta_n &\leq \frac{1}{n^2}(d + \frac{d^2}{2n^2} + \dots), \end{aligned}$$

for $|\frac{d}{n^2}| \leq \frac{1}{2}$, the series in the right-hand side of the inequality is convergent, and therefore

$$\begin{aligned} \delta_n &\leq \frac{e}{n^2} \\ \implies \delta_n &= O(n^{-2}). \end{aligned}$$

Therefore it is proved that, $\sum_{n=2}^{\infty} |\delta_n| < \infty$ by comparison test. Hence,

$$\lim_{n \rightarrow \infty} b_n = C_0 \text{ for some } C_0 \in \mathbb{R}.$$

Now, we will prove that $C_0 = \sqrt{2\pi}$. Consider the integral,

$$A_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx.$$

By integration by parts, we obtain that,

$$A_n = \frac{n-1}{n} A_{n-2}, \quad \text{where } n \geq 2.$$

Hence, for each positive integer $n \geq 1$,

$$\begin{aligned} A_{2n} &= \prod_{k=1}^n \frac{2k-1}{2k} \int_0^{\frac{\pi}{2}} 1 \, dx = \frac{\pi}{2} \prod_{k=1}^n \frac{2k-1}{2k}, \\ A_{2n+1} &= \prod_{k=1}^n \frac{2k}{2k+1} \int_0^{\frac{\pi}{2}} \sin x \, dx = \prod_{k=1}^n \frac{2k}{2k+1}. \end{aligned}$$

Now, we know that $A_{2n+1} \leq A_{2n} \leq A_{2n-1}$. Therefore, $\frac{A_{2n+1}}{A_{2n}} \leq 1$. Also, we have,

$$\frac{A_{2n+1}}{A_{2n}} = \frac{2n}{2n+1} \frac{A_{2n-1}}{A_{2n}} \geq \frac{2n}{2n+1}.$$

Therefore, by Squeeze theorem, we can say that,

$$\lim_{n \rightarrow \infty} \frac{A_{2n+1}}{A_{2n}} = 1.$$

i.e. we can deduce from here that,

$$\prod_{k=1}^{\infty} \frac{2k}{2k-1} \frac{2k}{2k+1} = \frac{\pi}{2}.$$

Also note that,

$$\prod_{k=1}^n \left(\frac{2k}{2k-1} \frac{2k}{2k+1} \right) = \frac{16^n (n!)^4}{(2n+1)(2n!)^2}.$$

We have already proved that $\lim_{n \rightarrow \infty} \frac{n!}{n^{(n+\frac{1}{2})} e^{-n}}$ exists and say its value is C . Now,

$$\lim_{n \rightarrow \infty} \frac{16^n (n!)^4}{(2n+1)(2n!)^2} = \frac{C^2}{4}.$$

So, it follows $C = \sqrt{2\pi}$, since $\frac{\pi}{2} = \lim_{n \rightarrow \infty} \frac{16^n (n!)^4}{(2n+1)(2n!)^2} = \frac{C^2}{4}$. □

Probability of returning to the origin

We now consider the number of times that the random walker returns to the origin. Here we use *indicator function* notation: If E is an event, then $1(E)$ is the random variable that takes the value 1 if the event occurs and 0 if the event does not occur. So, let $J_n = 1\{S_n = 0\}$. The total number of visits to the origin by the random walker is

$$V = \sum_{n=0}^{\infty} J_{2n}.$$

Note that

$$\mathbb{E}[V] = \sum_{n=0}^{\infty} \mathbb{E}[J_{2n}] = \sum_{n=0}^{\infty} \mathbb{P}\{S_{2n} = 0\}.$$

Applying Stirling's formula, $\mathbb{P}\{S_{2n} = 0\} \sim \frac{c}{\sqrt{n}}$ as $n \rightarrow \infty$. Therefore,

$$\mathbb{E}[V] = \infty.$$

Theorem 1. *The probability that a (one-dimensional) simple random walker returns to the origin infinitely often is one.*

Proof: Let q be the probability that the random walker ever returns to the origin after time 0. We will show that $q = 1$ by contradiction. Let if possible $q < 1$. Then we can give the distribution for V . Now, $\mathbb{P}\{V = 1\} = 1 - q$ since $V = 1$ if and only if the walker never returns after time zero. More generally,

$$\mathbb{P}\{V = k\} = q^{k-1}(1 - q), \text{ where } k = 1, 2, \dots$$

This tells us that

$$\mathbb{E}[V] = \sum_{k=1}^{\infty} k q^{k-1}(1 - q) = \frac{1}{1-q} < \infty.$$

But we know that $\mathbb{E}[V] = \infty$. Hence, it must be the case that $q = 1$. (proved)

5.2 Random walk in higher dimensions

We now consider the random walker on the d -dimensional integer grid,

$$\mathbb{Z}^d = \{(x_1, x_2, \dots, x_d) \mid x_j \in \mathbb{Z} \ \forall j \in \{1, 2, \dots, d\}\}$$

At each time step, the walker chooses one of its $2d$ nearest neighbours with probability $1/2d$, and moves to that site. We again let

$$S_n = x + X_1 + X_2 + \dots + X_n,$$

denotes the position of the particle. Here $x, X_1, \dots, X_n, S_n$ represent points in $\mathbb{Z}^d$, i.e. they are d -dimensional vectors with integer components. The increments $X_1, X_2, \dots, X_n$ are unit vectors with one component of absolute value 1 and the other components are zero. Again, we consider that $x = \mathbf{0}$.

Expectation

For $j \in \{1, 2, \dots, d\}, i \in \{1, 2, \dots, n\}$, we have,

$$\begin{aligned} \mathbb{E}(X_i^j) &= 1 \cdot \mathbb{P}(X_i^j = 1) + (-1) \cdot \mathbb{P}(X_i^j = -1) + 0 \cdot \mathbb{P}(X_i^j = 0) \\ &= 1 \cdot \mathbb{P}(X_i^j = 1, X_i^k = 0, \forall k \in \{1, 2, \dots, d\} \ \& \ k \neq j) + \\ &\quad (-1) \cdot \mathbb{P}(X_i^j = -1, X_i^k = 0, \forall k \in \{1, 2, \dots, d\} \ \& \ k \neq j) \\ &= 1 \cdot \frac{1}{2d} + (-1) \cdot \frac{1}{2d} \\ &= 0. \end{aligned}$$

Therefore, we can say that,

$$\begin{aligned} \mathbb{E}(S_n) &= (\mathbb{E}(S_n^1), \mathbb{E}(S_n^2), \dots, \mathbb{E}(S_n^d)) \\ &= (\mathbb{E}(X_1^1 + \dots + X_n^1), \dots, \mathbb{E}(X_1^d + \dots + X_n^d)) \\ &= (0, 0, \dots, 0), \end{aligned}$$

where the last equality holds by the linearity property of expectation.

Now, let us compute $\mathbb{E}(|S_n|^2)$. So,

$$\begin{aligned} \mathbb{E}(|S_n|^2) &= \mathbb{E}(S_n \cdot S_n) \\ &= \mathbb{E}\left(\sum_{i=1}^n X_i \cdot \sum_{i=1}^n X_i\right) \\ &= \mathbb{E}\left(\sum_{i=1}^n |X_i|^2 + \sum_{\substack{j=1 \\ j \neq k}}^n \sum_{k=1}^n X_j \cdot X_k\right) \\ &= \mathbb{E}\left(\sum_{i=1}^n |X_i|^2\right) + \mathbb{E}\left(\sum_{\substack{j=1 \\ j \neq k}}^n \sum_{k=1}^n X_j \cdot X_k\right) \end{aligned}$$

where the last equality holds by linearity property of expectation.

Now, $|X_i|^2 = 1 \forall i \in \{1, 2, \dots, n\}$.so, $\mathbb{E} \left(\sum_{i=1}^n |X_i|^2 \right) = n$, as $\mathbb{E}(c) = c$, if c is a constant. So,

$$\mathbb{E}(|S_n|^2) = n + \sum_{\substack{j=1 \\ j \neq k}}^n \sum_{k=1}^n \mathbb{E}(X_j \cdot X_k),$$

where the above equality holds for linearity property of expectation. Now, if $j \neq k$, $X_j \cdot X_k$ takes values 1, -1 and 0. Therefore,

$$\begin{aligned} \mathbb{E}(X_j \cdot X_k) &= \sum_x x \mathbb{P}(X_j \cdot X_k = x) \\ &= 1 \cdot \mathbb{P}(X_j \cdot X_k = 1) + (-1) \cdot \mathbb{P}(X_j \cdot X_k = -1) + 0 \cdot \mathbb{P}(X_j \cdot X_k = 0) \end{aligned}$$

Now, consider two events A_i, B_i ($i \in \mathbb{N}$) such that

$$\begin{aligned} A_i &= \{\omega | X_j^i(\omega) = X_k^i(\omega) = 1, X_j^l(\omega) = X_k^l(\omega) = 0 \forall l \neq i, l \in \{1, 2, \dots, d\}\}, \\ B_i &= \{\omega | X_j^i(\omega) = X_k^i(\omega) = -1, X_j^l(\omega) = X_k^l(\omega) = 0 \forall l \neq i, l \in \{1, 2, \dots, d\}\}. \end{aligned}$$

Then, we get,

$$\begin{aligned} \mathbb{P}(X_j \cdot X_k = 1) &= \mathbb{P} \left(\bigcup_{i=1}^d (A_i \cup B_i) \right) \\ &= \sum_{i=1}^d \mathbb{P}(A_i \cup B_i) \\ &= \sum_{i=1}^d (\mathbb{P}(A_i) + \mathbb{P}(B_i)) \\ &= \sum_{i=1}^d \left(\frac{1}{4d^2} + \frac{1}{4d^2} \right) \\ &= \frac{1}{2d}, \end{aligned}$$

where the second and third equality holds as the events are mutually exclusive, and the fifth equality holds for the independence of the random variables.

Probability of being at origin

In order to be at the origin, we need to take an even number of steps in each of the d -directions. Given that each of these numbers is even, the probability that each individual component is at the origin is the probability that a one dimensional walk is at the origin at time $2n/d$ (or, more precisely, an even integer very close to $2n/d$). Using this idea we get the asymptotics,

$$\mathbb{P}(S_{2n} = \mathbf{0}) \sim \frac{d^{d/2}}{\pi^{d/2} 2^{d-1} n^{d/2}}.$$

Note that, the fact that the exponent of n is $d/2$ is very important in this case.

Consider the expected number of returns to the origin. If V is the number of visits to the origin, then just as in the $d = 1$ case,

$$\mathbb{E}[V] = \sum_{n=0}^{\infty} \mathbb{P}\{S_{2n} = \mathbf{0}\}.$$

Also, $\mathbb{E}[V] = \frac{1}{1-q}$, where, $q = q_d$ is the probability that the d -dimensional walk returns to the origin. Since $\mathbb{P}\{S_{2n} = \mathbf{0}\} \sim \frac{d^{d/2}}{\pi^{d/2} 2^{d-1} n^{d/2}}$,

$$\mathbb{E}[V] = \sum_{n=0}^{\infty} \mathbb{P}\{S_{2n} = \mathbf{0}\} = \begin{cases} < \infty, & \text{if } d \geq 3 \\ = \infty, & \text{if } d = 2 \end{cases}$$

From the whole previous discussion, we state a theorem in the following.

Theorem 2. *Suppose S_n is simple random walk in $\mathbb{Z}^d$ with $S_0 = 0$. If $d = 1, 2$, the random walk is recurrent, i.e., with probability one it returns to the origin infinitely often. If $d \geq 3$, the random walk is transient, i.e., with probability one it returns to the origin only finitely often. Also,*

$$\mathbb{P}\{S_{2n} \neq 0 \text{ for all } n > 0\} > 0 \text{ if } d \geq 3.$$

Now, We do want to discuss one important lemma that probabilists use all the time. It is very easy but it has a name. (It is very common for mathematicians to assign names to lemmas that are used frequently if they are very simple—this way one can refer to them easily.)

Lemma 1. (Borel-Cantelli Lemma). *Suppose $E_1, E_2, \dots$ is a collection of events such that*

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty.$$

Then with probability one at most finitely many of the events occur.

As an example, consider the simple random walk in $\mathbb{Z}^d$, $d \geq 3$ and let E_n be the event that $S_n = 0$. Then, the estimates from this section show that

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty,$$

and hence with probability one, only finitely many of the events E_n occur. This says that with probability one, the random walk visits the origin only finitely often.

5.3 Boundary value problems

One dimension: gambler's ruin

Consider the random walk with absorbing boundary on $\{0, \dots, N\}$. Let $\alpha(j) = \alpha(j, N)$ be the probability that the walker starting at state j eventually ends up absorbed in state N . We can give a gambling interpretation to this by viewing $\alpha(j)$ as the number of chips currently held by a gambler who is playing a fair game where at each time duration the player wins or loses one chip. The gambler starts with j chips and plays until he or she has N chips or has gone bankrupt. Clearly, $\alpha(0) = 0$, $\alpha(N) = 1$.

Suppose $0 < j < N$. If we consider that the gambler has either $j + 1$ chips or $j - 1$ chips after the first game with equal probability, then

$$\alpha(j) = \frac{1}{2}\alpha(j+1) + \frac{1}{2}\alpha(j-1). \quad (5.1)$$

One function that satisfies the conditions discussed above, is the linear function $\alpha(j) = j/N$. In fact, this is the only function that we will now prove.

Theorem 3. *Suppose a, b are real numbers and N is a positive integer. Then the only function $F : \{0, \dots, N\} \rightarrow \mathbb{R}$ satisfying (5.1) with $F(0) = a$ and $F(N) = b$ is the linear function*

$$F_0(x) = a + \frac{x(b-a)}{N}.$$

Proof. Suppose that F is a solution to (5.1). Then for each $0 < x < N$,

$$F(x) \leq \max\{F(x-1), F(x+1)\}.$$

Using this, we can say that the maximum and minimum of F is obtained on $\{0, N\}$. Now, suppose that $F(0) = F(N) = 0$. Then, the maximum and minimum of the functions are both zero. i.e.

$$0 \leq F(x) \leq 0 \quad \forall \quad x \in \{0, 1, \dots, N\},$$

hence, $F \equiv 0$. Let $F(0) = a$, $F(N) = b$ and let F_0 be the linear function with these same boundary conditions. Then $F - F_0$ satisfies (5.1) with boundary value 0, hence is identically zero, i.e. $F - F_0 \equiv 0$. This implies that $F = F_0$. $\square$

Higher dimensions

We will generalize the previous result to higher dimensions. We replace the interval $\{1, \dots, N\}$ with an arbitrary finite subset A of $\mathbb{Z}^d$. Let ∂A be the (outer) boundary of A defined by

$$\partial A = \{z \in \mathbb{Z}^d : \text{dist}(z, A) = 1\},$$

and we let $\bar{A} = A \cup \partial A$ be the closure of A .

We define two linear operators $\mathbf{Q}$ and $\mathcal{L}$ on functions by

$$\begin{aligned} \mathbf{Q}F(x) &= \frac{1}{2d} \sum_{y \in \mathbb{Z}^d, |x-y|=1} F(y), \\ \mathcal{L}F(x) &= (\mathbf{Q} - \mathbf{I})F(x) = \frac{1}{2d} \sum_{y \in \mathbb{Z}^d, |x-y|=1} [F(y) - F(x)]. \end{aligned}$$

The operator $\mathcal{L}$ is often called the (discrete) Laplacian. Now, we let $T_A = \min\{n \geq 0 : S_n \notin A\}$. Then we state the generalized version of the previous theorem as follows.

Theorem 4. *If $A \subset \mathbb{Z}^d$ is finite, then for every $F : \partial A \rightarrow \mathbb{R}$, there is a unique extension F to $\bar{A}$ that satisfies the equation $\mathcal{L}F(x) = 0$. It is given by*

$$F_0(x) = \mathbb{E}[F(S_{T_A}) | S_0 = x] = \sum_{y \in \partial A} \mathbb{P}\{S_{T_A} = y | S_0 = x\} F(y).$$

5.4 Heat equation

Formation of discrete heat equation

We will now introduce a mathematical model for heat flow. The probabilistic view of this model is given by imagining that the temperature in A is controlled by a very large number of heat particles which perform random walks on A until they leave A at which time they are killed. The temperature at x , at time n , $p_n(x)$ is given by the density of the particles at x . For $x \in A$, the temperature at x is given by,

$$p_{n+1}(x) = \frac{1}{2d} \sum_{|y-x|=1} p_n(y).$$

If we introduce the notation $\partial_n p_n(x) = p_{n+1}(x) - p_n(x)$, we get the *heat equation*

$$\partial_n p_n(x) = \mathcal{L}p_n(x), \quad x \in A, \quad (5.2)$$

where $\mathcal{L}$ denotes the discrete Laplacian as before. The initial temperature is given as an initial condition

$$p_0(x) = f(x), \quad x \in A. \quad (5.3)$$

We write the boundary condition as,

$$p_n(x) = 0, \quad x \in \partial A. \quad (5.4)$$

If $x \in A$ and the initial condition is $f(x) = 1$ and $f(z) = 0$ for $z \neq x$, then

$$p_n(y) = \mathbb{P}\{S_{n \wedge T_A} = y \mid S_0 = x\}.$$

Solution of discrete heat equation

The solution of the discrete heat equation can be found by the *separation of variables* technique but it may not always be possible to give nice expressions for the eigenvalues and eigenvectors. We summarize here.

Theorem 5. *If A is a finite subset of $\mathbb{Z}^d$ with N elements, then we can find N linearly independent eigenfunctions $\phi_1, \phi_2, \dots, \phi_N$ of the operator $\mathbf{Q}$ with real eigenvalues $\lambda_1, \dots, \lambda_N$. The solution to (5.2)-(5.4) is given by*

$$p_n(x) = \sum_{j=1}^N c_j \lambda_j^n \phi_j(x),$$

where c_j are chosen so that

$$f(x) = \sum_{j=1}^N c_j \phi_j(x).$$

In fact, the ϕ_j can be chosen to be orthonormal,

$$\langle \phi_j, \phi_k \rangle := \sum_{x \in A} \phi_j(x) \phi_k(x) = \delta(k - j).$$

(Here we have introduced the delta function notation, $\delta(z) = 1$ if $z = 0$ and $\delta(z) = 0$ if $z \neq 0$.)

We will write $p_n(x, y; A)$ to be the solution of the heat equation with initial condition equal to one at x and 0 otherwise. In other words,

$$p_n(x, y; A) = \mathbb{P}\{S_n = y, T_A > n \mid S_0 = x\}, \quad x, y \in A.$$

Green's function

For any $y \in A$, let V_y denote the number of visits to y before leaving A ,

$$V_y = \sum_{n=0}^{T_A-1} 1\{S_n = y\} = \sum_{n=0}^{\infty} 1\{S_n = y, T_A > n\}.$$

Here, we again use the indicator function notation. Note that

$$\mathbb{E}[V_y \mid S_0 = x] = \sum_{n=0}^{\infty} \mathbb{P}\{S_n = y, T_A > n \mid S_0 = x\} = \sum_{n=0}^{\infty} p_n(x, y; A).$$

This quantity is of sufficient interest that it is given a name. The *Green's function* $G_A(x, y)$ is the function on $A \times A$ given by

$$G_A(x, y) = \mathbb{E}[V_y \mid S_0 = x] = \sum_{n=0}^{\infty} p_n(x, y; A).$$

We define $G_A(x, y) = 0$ if $x \notin A$ or $y \notin A$. From this, we summarize the main result as follows.

Theorem 6. *Suppose A is a bounded subset of $\mathbb{Z}^d$, and $g : A \rightarrow \mathbb{R}$ is a given function. Then the unique function $F : \bar{A} \rightarrow \mathbb{R}$ satisfying*

$$\begin{aligned} F(x) &= 0, & x \in \partial A, \\ \mathcal{L}F(x) &= -g(x), & x \in A, \end{aligned}$$

is

$$F(x) = \mathbb{E} \left[\sum_{j=0}^{T_A-1} g(S_j) \mid S_0 = x \right] = \sum_{y \in A} g(y) G_A(x, y).$$

Chapter 6

Conclusion & Suggestion of Research Project

6.1 Conclusion

In this project work, our primary objective was to try to find the connection between the random walk and the heat equation i.e. to make a bridge between two different subjects, probability theory and partial differential equations. We saw how a discrete random process random walk can be related to a continuous deterministic process heat equation by the formation of the latter in a different version. Physically, this relationship demonstrates how the random movement of particles on a microscopic scale leads to the macroscopic phenomenon of heat conduction. The broader significance of this study lies in its potential applications across multiple disciplines. Understanding the connection between random walks and the heat equation can enhance modeling techniques in various fields like physics, biology, finance, computer science etc.

6.2 Suggestion

The project work on the topic of random walk and heat equation demands some suggestions regarding its domain. Further research could examine the connection between random walks and stochastic differential equations, which generalize the heat equation by incorporating random noise. This approach could offer a more comprehensive understanding of complex systems influenced by randomness. Investigating its applications in various disciplines like biology, finance, physics, computer science etc will enhance its utility in future research.

Chapter 7

Results Achieved

In this section, we will summarize the results obtained from the whole discussion on random walk and heat equation. First, we will discuss one dimensional random walk, then uplift the idea to higher dimensions and finally the heat equation.

7.1 One dimensional random walk

We analyzed a simple random walk on the integers, where the walker moves left or right with equal probability at each step. We derived the expectation and variance of the position of the walker after n steps, showing that the expectation is zero and the variance is n . Then We computed the probability of the walker being at a particular position after $2n$ steps. Then we deduce Stirling's formula to see the behaviour of $n!$. Finally, We proved that the walker in one dimension returns to to origin infinitely often.

7.2 Higher dimensional random walk

Then we increase the dimension to higher ones, where the walker has many choices to walk with equal probability. Then we compute the expectation of various random variables and prove that for higher dimensions, the walk is transient to the origin. i.e. the walker returns to the origin only finitely often.

7.3 Heat equation

Lastly, we introduce the heat equation in a different way from the probabilistic point of view where the particles perform random walks. Then we deduce the solution and define the Green's function and finally connect the two topics.

Chapter 8

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